07/08/26

ITA0512 - COMPUTER VISION

CO2 - AT3 - CLASS TEST

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1) An image requires enhancement of edges without amplifying noise significantly. Explain and justify a suitable spatial or frequency domain method.

The Laplacian of Gaussian (LoG) or unsharp Masking method is suitable for enhancing edges while minimizing noise.

* First, a Gaussian filter smooths the image to reduce noise.
* Then, the LoG operator detects edges.
* combining smoothing and edge detection prevents excessive noise amplification.
* This method produces sharper edges while maintaining image quality.

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Expected Result: clear and enhanced edges with minimal noise.

2) A grayscale image shows poor sepration between object and background intensities. Describe and justify a thresholding approach.

Otsu's Thresholding is an effective method for automatic image segmentation.

* It automatically selects the optimal threshold value.

* It separates the image into foreground and background based on pixel values.

* It minimizes the variance within each class.

* No manual threshold selection is required.

Expected Result :- Improved segmentation with clear separation between the object and the background.

3) An image contains multiple regions with similar intensity but different textures. Explain and justify a region-based segmentation approach.

Region growing is a suitable region based segmentation technique.

* It starts with one or more seed pixels.

* Neighboring pixels with similar properties are grouped together.

* The process continues until no more similar pixels are found.

* It effectively separates regions based on similarity rather than just intensity.

Expected Result :- Accurate grouping of textured regions into meaningful segments.

4) Detected edges are noisy and fragmented. Describe and justify a method to refine edges and improve boundary detection.

The Canny Edge Detection algorithm is the most suitable method.

* It smooths the image using a Gaussian filter.
* Computes gradient magnitude and direction.
* Applies non-maximum supression to obtain thin edges.
* Uses double thresholding and edge tracking to remove weak and noisy edges.

Expected Outcome :- continuous, accurate, and noise free edges suitable for segmentation.

5) An application requires precise object boundary extraction for shape analysis. Explain and justify a suitable segmentation and boundary detection approach.

Watershed segmentation combined with Canny Edge Detection is an effective approach.

* Watershed segmentation separates touching or overlapping objects.
* Canny edge detection accurately extracts object boundaries.

* The combination provides precise segmentation and boundary extraction.

* It is widely used in medical imaging, industrial inspection, and object recognition.

Expected Result :- Accurate object boundaries that are suitable for shape analysis and feature extraction.